

Ádám Fodor

MACHINE LEARNING ENGINEER

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PROFILE

A decade of building and shipping ML solutions: taking multimodal deep-learning models from design through evaluation, optimization and deployment, with ongoing maintenance as versioned packages. **Nine PyPI packages**, two deployed services with public demos, and ten peer-reviewed publications, including **two Q1 journal papers**.

EXPERIENCE

Research Scientist, Neural Information Processing Group (NIPG) Sep 2017 – Present
Eötvös Loránd University, Faculty of Informatics, Dept. of Artificial Intelligence · Budapest, Hungary

- **Designed LinMult**, a general-purpose multimodal sequence model with **linear-complexity attention**, then adapted it across three projects, each evaluated **cross-dataset**: blink **event detection**; personality and behavior estimation; emotion estimation with **multi-task learning across disjoint datasets**.
- **Shipped an eye blink detector end-to-end**: model, **Q2** journal paper, PyPI package, real-time inference. Then ported it to *Tobii* head-mounted camera video and integrated it into an ADOS-2 autism behavior-analysis framework delivered to **Rush University Medical Center** and published in a **Q1** journal.
- **Split the runtime, PyTorch for development and ONNX for release**: the blink detector's released package drops torch entirely, 469 MB smaller and 4.15 ms/frame instead of 20.19; every export is gated on numerical parity probes against the *PyTorch* graph so the two paths do not drift apart.
- **Put the personality model on Google Cloud Run**: *MLflow registry* as the promotion boundary, Optuna hyperparameter search, ONNX model served from Docker, scale-to-zero so an idle demo costs nothing.
- **Back metric gains with significance tests**: paired bootstrap, 95% CIs, Holm-corrected p-values, so an improvement that sits inside the noise does not get reported as a win.
- **Maintain nine PyPI packages at 84–100% coverage**, CI-enforced via GitHub Actions and Codecov; **four** of the group's projects are built on the *exordium* feature-extraction toolkit.
- **Implemented the video processing and object tracking** under project EFOP-3.6.3-VEKOP-2017-00002, published as **two IJCNN (CORE A)** papers.

Associate Scientist, Institute for Computer Science and Control (SZTAKI) Jul 2014 – Aug 2017
Hungarian Academy of Sciences · Budapest, Hungary

- **Developed and published AHFS**, an adaptive hybrid feature selection method, in **Pattern Recognition (Q1)**: on the paper's benchmark it roughly **halves the model error** of the individual selectors it combines, on industrial data up to 1,100 variables wide. Deployed on wind-turbine monitoring and a machining situation-detection model that **runs in the machines' control systems**; since reused in clinical EEG and actigraphy screening, where it was the **best-performing** selector at 0.89–0.92 accuracy.
- **Ran a basic research project for OPEL Szentgotthárd** under iKOMP: feature selection and production trend forecasting on shop-floor data, which the plant used to bring engine failures down.
- **Built the institute's model-training and visualization framework**, extending its earlier tooling with data mining support; **SZTAKI adopted it** as the successor to their in-house system.

EDUCATION

PhD in Computer Science, summa cum laude Sep 2019 – Jun 2026
Eötvös Loránd University (ELTE)

Thesis: Affective Behavior and Social Interaction Analysis
Supervisor: Dr. habil. András Lőrincz

MSc in Computer Science, excellent Feb 2016 – Jul 2018
Eötvös Loránd University (ELTE)

Thesis: Speech de-identification with deep neural networks
Supervisor: Dr. habil. András Lőrincz

BSc in Computer Science, excellent Sep 2012 – Feb 2016
Eötvös Loránd University (ELTE)

Thesis: User interface design and implementation for data mining algorithms in MATLAB
Supervisor: Dr. Zsolt János Viharos

SKILLS

ML & Deep Learning:	PyTorch, Lightning, Transformers, computer vision, multimodal learning, multi-task learning, distributed training (multi-GPU)
GenAI & Agentic AI:	LangChain, LangGraph, Ollama, local LLMs, agentic workflows
ML Systems & Optimization:	ONNX Runtime, model quantization & optimization, pruning, inference optimization
MLOps & Deployment:	MLflow, Optuna, Docker, Google Cloud Run, FastAPI, Gradio, Hugging Face Hub & Spaces, GitHub Actions, Codecov
Data & Pipelines:	NumPy, pandas, SQLAlchemy, HDF5, Hugging Face Datasets, video decoding, reproducible data pipelines, dataset unification & balancing
Engineering:	Python, Bash, Linux, Git, uv, Make, test-driven development, mypy, ruff, semantic versioning, Sphinx
Evaluation:	experimental design, hypothesis testing, bootstrap confidence intervals, multiple-comparison correction, ablation studies

SELECTED PROJECTS

- BlinkLinMult** · Transformer-based eye blink detection and eye state recognition.
End-to-end PyTorch-to-ONNX Runtime CV pipeline with numerical-parity checks; reduced latency from 94 to 23 ms/frame (42.8 fps). REST API, Docker and live demo.
PyTorch · ONNX Runtime · FastAPI · Docker · Gradio · Hugging Face Spaces
- PersonalityLinMult** · Multimodal Big Five personality perception.
Acoustic, visual and text model optimized for CCC to avoid mean-prediction collapse; ONNX export provided 6-8× faster inference with ≤ 0.004 trait-level drift. Tracked and tuned with MLflow and Optuna; deployed on GCR.
PyTorch · MLflow · Optuna · ONNX Runtime · FastAPI · Docker · Cloud Run
- Annie** · Local-first video annotation and dataset curation tool.
Human-in-the-loop review, correction and event annotation for video detections, tracks and temporal segments; privacy-preserving local operation. Released on PyPI and Docker Hub.
PyTorch · NiceGUI · torchcodec · SQLite · Docker · GitHub Actions
- labmate** · Local assistant for research and portfolio workflows.
Agentic, source-grounded local LLM workflows for paper analysis, literature research and portfolio updates. Combines retrieval, controlled tool calls, parallel extraction and human approval; evaluator checks generated claims against source excerpts.
Python · LangGraph · Ollama · Qwen · Gemma · Pydantic · Graphviz

SELECTED PUBLICATIONS

- B. C. Dos Santos Melício, K. Karaköse, Á. Fodor, L. Xiang, V. Varga, L. Soorya, E. Dillon, P. Kun, A. Sárkány, M. Chetouani, K. Fenech. **Multimodal Framework for Automatic Behavior Analysis of Children with Autism During ADOS-2**. *Cognitive Computation*, vol. 17, 137, 2025. (Q1)
- Á. Fodor, K. Fenech, A. Lőrincz. **BlinkLinMult: Transformer-Based Eye Blink Detection**. *MDPI Journal of Imaging*, vol. 9-10, 196, 2023. (Q2)
- Z. J. Viharos, K. B. Kis, Á. Fodor, M. I. Büki. **Adaptive, hybrid feature selection (AHFS)**. *Pattern Recognition*, vol. 116, 107932, 2021. (Q1)

Full list of ten publications at adamfodor.com

ADDITIONAL INFORMATION

Awards:	1st place, Students' Scientific Conference (TDK), 2018 Special award, Morgan Stanley, 2018 Student TDK supervision award, 2023
Mentoring:	Supervised 2 BSc/MSc research projects and taught MSc courses over 8 semesters.
Languages:	Hungarian: native English: fluent
References:	Dr. habil. András Lőrincz · Founder of NIPG, ELTE · lorincz@inf.elte.hu Dr. Kristian Fenech · Head of NIPG, ELTE · fenech@inf.elte.hu